Ex5: Information Extraction using spaCy

Dataset: Amazon Fine Food Reviews (Kaggle) – `Reviews.csv` (same dataset as Ex4)

The dataset (~300 MB) is downloaded automatically from Kaggle the first time you run the notebook. If you already have `Reviews.csv`, upload it to the Colab **Files** panel and it is used instead.

Task 1: Install and import the required libraries, load the dataset

```
!pip install -q spacy
!python -m spacy download en_core_web_sm
```

```
import os
import time
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from collections import Counter
import spacy

sns.set_style('whitegrid')
pd.set_option('display.max_colwidth', 150)

nlp = spacy.load("en_core_web_sm")
print("spaCy version:", spacy.__version__)
print("Pipeline components:", nlp.pipe_names)
```

```
spaCy version: 3.8.16
Pipeline components: ['tok2vec', 'tagger', 'parser', 'attribute_ruler', 'lemmatizer', 'ner']
```

```
file_name = "Reviews.csv"
if not os.path.exists(file_name):  # file not uploaded -> download from Kaggle
    import kagglehub
    path = kagglehub.dataset_download("snap/amazon-fine-food-reviews")
    file_name = os.path.join(path, "Reviews.csv")

df = pd.read_csv(file_name)
print("Shape of the dataset:", df.shape)
display(df.head())
```

```
Shape of the dataset: (568454, 10)
```

	Id	ProductId	UserId	ProfileName	HelpfulnessNumerator	HelpfulnessDenominator	Score	Time	Summary
0	1	B001E4KFG0	A3SGXH7AUHU8GW	delmartian	1	1	5	1303862400	Good Quality Dog Food
1	2	B00813GRG4	A1D87F6ZCVE5NK	dll pa	0	0	1	1346976000	Not as Advertised
2	3	B000LQOCH0	ABXLMWJIXXAIN	Natalia Corres "Natalia Corres"	1	1	4	1219017600	"Delight" says it all
3	4	B000UA0QIQ	A395BORC6FGVXV	Karl	3	3	2	1307923200	Cough Medicine
4	5	B006K2ZZ7K	A1UQRSCLF8GW1T	Michael D. Bigham "M. Wassir"	0	0	5	1350777600	Great taffy

	Text
0	I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew t...
1	Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this was an error or if the vendor i...
2	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut...
3	If you are looking for the secret ingredient in Robitussin I believe I have found it. I got this in addition to the Root Beer Extract I ordered (...
4	Great taffy at a great price. There was a wide assortment of yummy taffy. Delivery was very quick. If your a taffy lover, this is a deal.

Task 2: Clean the dataset

Select the `Text` column, remove missing values, remove duplicate reviews, keep the first 10,000 reviews.

```
reviews = df[['Text']].copy()

print("Missing values in Text:", reviews['Text'].isnull().sum())
reviews = reviews.dropna(subset=['Text'])
```

```
print("Duplicate reviews:", reviews.duplicated(subset='Text').sum())
reviews = reviews.drop_duplicates(subset='Text')

reviews = reviews.head(10000).reset_index(drop=True)

# remove HTML line-break tags (<br />) and extra spaces so they are not treated as tokens
reviews['Text'] = (reviews['Text'].astype(str)
                   .str.replace(r'<.*?>', ' ', regex=True)
                   .str.replace(r'\s+', ' ', regex=True)
                   .str.strip())

print("Shape after cleaning:", reviews.shape)
display(reviews.head())
```

```
Missing values in Text: 0
Duplicate reviews: 174875
Shape after cleaning: (10000, 1)
```

	Text
0	I have bought several of the Vitality canned dog food products and have found them all to be of good quality. The product looks more like a stew t...
1	Product arrived labeled as Jumbo Salted Peanuts...the peanuts were actually small sized unsalted. Not sure if this was an error or if the vendor i...
2	This is a confection that has been around a few centuries. It is a light, pillowy citrus gelatin with nuts - in this case Filberts. And it is cut ...
3	If you are looking for the secret ingredient in Robitussin I believe I have found it. I got this in addition to the Root Beer Extract I ordered (w...
4	Great taffy at a great price. There was a wide assortment of yummy taffy. Delivery was very quick. If your a taffy lover, this is a deal.

Text Preprocessing and Part-of-Speech (POS) Tags

Task 3: Apply the spaCy pipeline and show tokens, lemmas and POS tags for five reviews

```
start = time.time()
docs = list(nlp.pipe(reviews['Text'], batch_size=256))
print(f"Processed {len(docs)} reviews with spaCy in {time.time() - start:.1f} seconds")
```

```
Processed 10000 reviews with spaCy in 67.5 seconds
```

```
# five short reviews (12-25 tokens) so the token tables are easy to read
sample_ids = [i for i, doc in enumerate(docs) if 12 <= len(doc) <= 25][:5]

for n, i in enumerate(sample_ids, start=1):
    doc = docs[i]
    print(f"Review {n} (row {i}): {doc.text}")
    token_table = pd.DataFrame([[token.text for token in doc],
                                [token.lemma_ for token in doc],
                                [token.pos_ for token in doc]],
                               index=['Token', 'Lemma', 'POS Tag'], columns=range(1, len(doc) + 1))
    display(token_table)
```

```
Review 1 (row 13): good flavor! these came securely packed... they were fresh and delicious! i love these Twizzlers!
```

	1	2	3	4	5	6	7	8	9
Token	good	flavor	!	these	came	securely	packed	...	they
Lemma	good	flavor	!	these	come	securely	packed	...	they
POS Tag	ADJ	NOUN	PUNCT	PRON	VERB	ADV	ADJ	PUNCT	PRON

	10	11	12	13	14	15	16	17	18
Token	were	fresh	and	delicious	!	i	love	these	Twizzlers
Lemma	be	fresh	and	delicious	!	I	love	these	Twizzlers
POS Tag	AUX	ADJ	CCONJ	ADJ	PUNCT	PRON	VERB	DET	PROPN

	19
Token	!
Lemma	!
POS Tag	PUNCT

```
Review 2 (row 14): The Strawberry Twizzlers are my guilty pleasure - yummy. Six pounds will be around for a while with my son and I.
```

	1	2	3	4	5	6	7	8	9
Token	The	Strawberry	Twizzlers	are	my	guilty	pleasure	-	yummy
Lemma	the	Strawberry	Twizzlers	be	my	guilty	pleasure	-	yummy
POS Tag	DET	PROPN	PROPN	AUX	PRON	ADJ	NOUN	PUNCT	ADJ

	10	11	12	13	14	15	16	17	18
Token	.	Six	pounds	will	be	around	for	a	while
Lemma	.	six	pound	will	be	around	for	a	while
POS Tag	PUNCT	NUM	NOUN	AUX	AUX	ADV	ADP	DET	NOUN

	19	20	21	22	23
Token	with	my	son	and	I.
Lemma	with	my	son	and	I.
POS Tag	ADP	PRON	NOUN	CCONJ	PROPN

```
Review 3 (row 23): I love this candy. After weight watchers I had to cut back but still have a craving for it.
```

	1	2	3	4	5	6	7	8	9
Token	I	love	this	candy	.	After	weight	watchers	I
Lemma	I	love	this	candy	.	after	weight	watcher	I
POS Tag	PRON	VERB	DET	NOUN	PUNCT	ADP	NOUN	NOUN	PRON

	10	11	12	13	14	15	16	17	18
Token	had	to	cut	back	but	still	have	a	craving
Lemma	have	to	cut	back	but	still	have	a	craving
POS Tag	VERB	PART	VERB	ADV	CCONJ	ADV	VERB	DET	NOUN

	19	20	21
Token	for	it	.
Lemma	for	it	.
POS Tag	ADP	PRON	PUNCT

```
Review 4 (row 25): Product received is as advertised. Twizzlers, Strawberry, 16-Ounce Bags (Pack of 6)
```

	1	2	3	4	5	6	7	8	9
Token	Product	received	is	as	advertised	.	Twizzlers	,	Strawberry
Lemma	product	receive	be	as	advertise	.	Twizzlers	,	Strawberry
POS Tag	NOUN	VERB	AUX	ADV	VERB	PUNCT	PROPN	PUNCT	PROPN

	10	11	12	13	14	15	16	17	18
Token	,	16	-	Ounce	Bags	(	Pack	of	6
Lemma	,	16	-	Ounce	Bags	(	Pack	of	6
POS Tag	PUNCT	NUM	PUNCT	PROPN	PROPN	PUNCT	PROPN	ADP	NUM

	19
Token	)
Lemma	)
POS Tag	PUNCT

```
Review 5 (row 26): The candy is just red , No flavor . Just plan and chewy . I would never buy them again
```

	1	2	3	4	5	6	7	8	9
Token	The	candy	is	just	red	,	No	flavor	.
Lemma	the	candy	be	just	red	,	no	flavor	.
POS Tag	DET	NOUN	AUX	ADV	ADJ	PUNCT	DET	NOUN	PUNCT

	10	11	12	13	14	15	16	17	18
Token	Just	plan	and	chewy	.	I	would	never	buy
Lemma	just	plan	and	chewy	.	I	would	never	buy
POS Tag	ADV	NOUN	CCONJ	NOUN	PUNCT	PRON	AUX	ADV	VERB

	19	20
Token	them	again
Lemma	they	again
POS Tag	PRON	ADV

Task 4: Identify and display POS categories – Nouns, Verbs, Adjectives, Adverbs

```
pos_categories = {'Nouns': 'NOUN', 'Verbs': 'VERB', 'Adjectives': 'ADJ', 'Adverbs': 'ADV'}

# POS categories in the same five sample reviews
rows = []
for n, i in enumerate(sample_ids, start=1):
    row = {'Review': n}
    for name, tag in pos_categories.items():
        row[name] = ', '.join(token.text for token in docs[i] if token.pos_ == tag)
```

```
    rows.append(row)
display(pd.DataFrame(rows).set_index('Review'))
```

Review	Nouns	Verbs	Adjectives	Adverbs
1	flavor	came, love	good, packed, fresh, delicious	securely
2	pleasure, pounds, while, son		guilty, yummy	around
3	candy, weight, watchers, craving	love, had, cut, have		back, still
4	Product	received, advertised		as
5	candy, flavor, plan, chewy	buy	red	just, Just, never, again

```
# Most common words in each POS category across all 10,000 reviews
category_counters = {name: Counter() for name in pos_categories}
for doc in docs:
    for token in doc:
        for name, tag in pos_categories.items():
            if token.pos_ == tag and token.is_alpha:
                category_counters[name][token.lemma_.lower()] += 1

top_words = {}
for name, counter in category_counters.items():
    words = [f"{w} ({c})" for w, c in counter.most_common(15)]
    top_words[name] = words + [''] * (15 - len(words))
top_words = pd.DataFrame(top_words, index=range(1, 16))
print("Top 15 words (lemma and count) in each POS category:")
display(top_words)
```

```
Top 15 words (lemma and count) in each POS category:
```

	Nouns	Verbs	Adjectives	Adverbs
1	flavor (3269)	have (6535)	good (4618)	so (3279)
2	product (3095)	make (3225)	great (2852)	just (2932)
3	coffee (2986)	use (2847)	other (1854)	very (2837)
4	taste (2149)	try (2779)	little (1372)	really (1752)
5	food (2039)	love (2740)	more (1332)	too (1593)
6	time (1722)	get (2645)	well (997)	well (1516)
7	tea (1702)	buy (2632)	sweet (873)	also (1468)
8	cup (1619)	find (2317)	hot (868)	as (1310)
9	bag (1560)	like (2292)	free (804)	even (1172)
10	price (1404)	taste (2201)	many (746)	only (1150)
11	chip (1300)	eat (1801)	small (734)	now (1082)
12	store (1097)	go (1567)	easy (712)	much (1028)
13	sugar (1096)	give (1342)	nice (699)	again (1017)
14	brand (1067)	think (1326)	healthy (688)	then (964)
15	water (1060)	order (1287)	bad (681)	more (898)

Task 5: Frequency of the major POS categories in the reviews

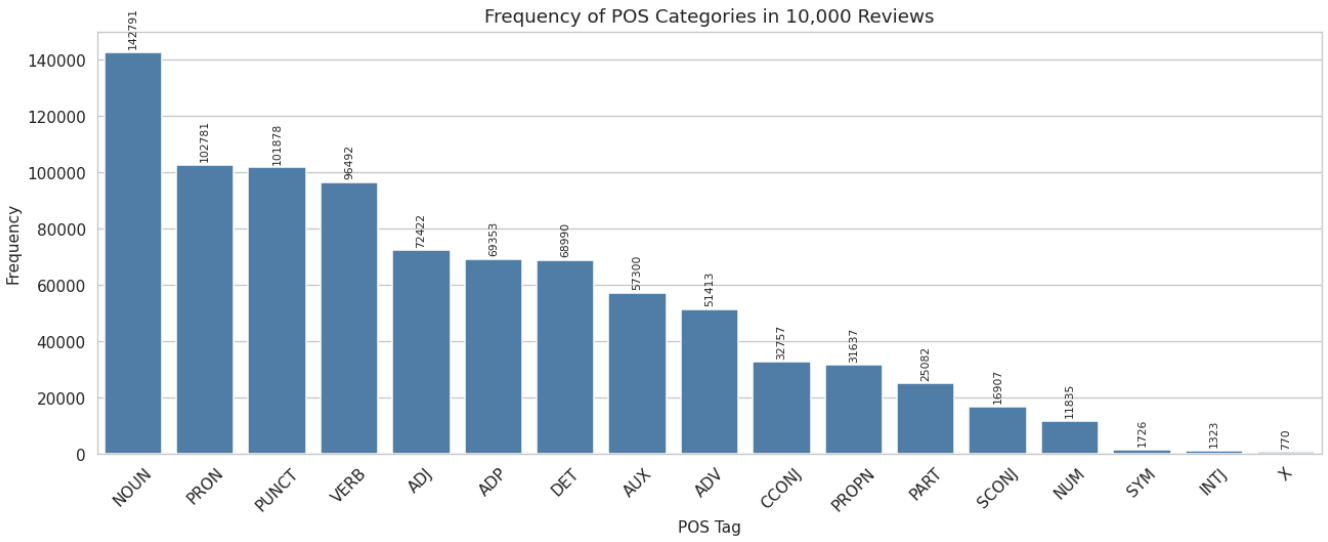
```
pos_counts = Counter(token.pos_ for doc in docs for token in doc if token.pos_ != 'SPACE')

pos_df = pd.DataFrame(pos_counts.most_common(), columns=['POS Tag', 'Frequency'])
pos_df['Description'] = pos_df['POS Tag'].apply(spacy.explain)
pos_df['Percentage (%)'] = (pos_df['Frequency'] / pos_df['Frequency'].sum() * 100).round(2)
display(pos_df)

plt.figure(figsize=(12, 5))
ax = sns.barplot(data=pos_df, x='POS Tag', y='Frequency', color='steelblue')
ax.bar_label(ax.containers[0], fontsize=7, rotation=90, padding=2)
plt.title('Frequency of POS Categories in 10,000 Reviews')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

	POS Tag	Frequency	Description	Percentage (%)
0	NOUN	142791	noun	16.13
1	PRON	102781	pronoun	11.61
2	PUNCT	101878	punctuation	11.51
3	VERB	96492	verb	10.90
4	ADJ	72422	adjective	8.18
5	ADP	69353	adposition	7.83
6	DET	68990	determiner	7.79
7	AUX	57300	auxiliary	6.47

	POS Tag	Frequency	Description	Percentage (%)
8	ADV	51413	adverb	5.81
9	CCONJ	32757	coordinating conjunction	3.70
10	PROPN	31637	proper noun	3.57
11	PART	25082	particle	2.83
12	SCONJ	16907	subordinating conjunction	1.91
13	NUM	11835	numeral	1.34
14	SYM	1726	symbol	0.19
15	INTJ	1323	interjection	0.15
16	X	770	other	0.09



Named Entity Recognition (NER)

Task 6: Extract PERSON, ORG, DATE, MONEY, QUANTITY, PRODUCT entities into a structured DataFrame

```
target_labels = ['PERSON', 'ORG', 'DATE', 'MONEY', 'QUANTITY', 'PRODUCT']

entity_rows = []
for review_id, doc in enumerate(docs):
    for ent in doc.ents:
        if ent.label_ in target_labels:
            entity_rows.append({'Review_ID': review_id, 'Entity': ent.text,
                                'Label': ent.label_, 'Description': spacy.explain(ent.label_)})

entities_df = pd.DataFrame(entity_rows)
print("Total entities extracted:", len(entities_df))
display(entities_df.head(20))
```

```
Total entities extracted: 16149
```

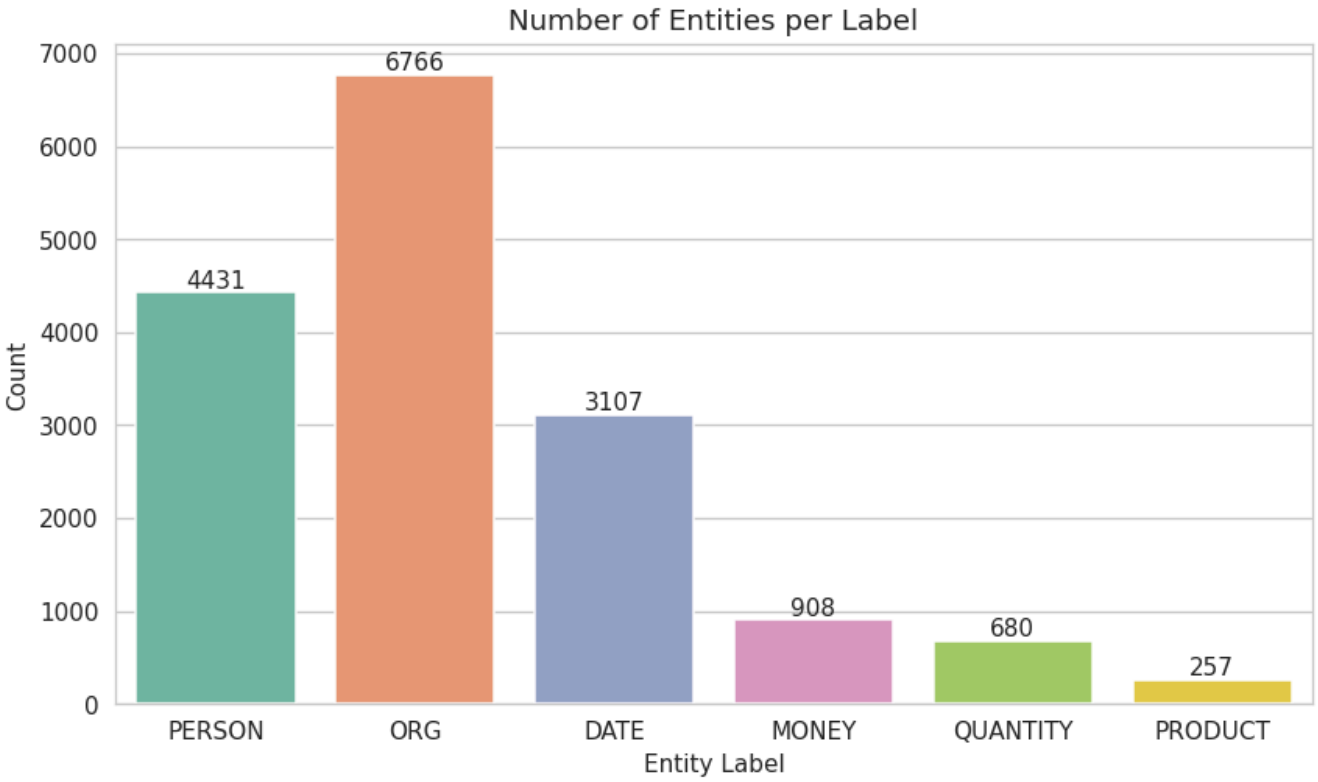
	Review_ID	Entity	Label	Description
0	0	Vitality	ORG	Companies, agencies, institutions, etc.
1	0	My Labrador	PERSON	People, including fictional
2	1	Jumbo Salted	ORG	Companies, agencies, institutions, etc.
3	2	around a few centuries	DATE	Absolute or relative dates or periods
4	2	pillowy citrus gelatin	ORG	Companies, agencies, institutions, etc.
5	2	C.S. Lewis'	ORG	Companies, agencies, institutions, etc.
6	2	Edmund	PRODUCT	Objects, vehicles, foods, etc. (not services)
7	2	Sisters	PERSON	People, including fictional
8	2	Witch	ORG	Companies, agencies, institutions, etc.
9	5	five pound	QUANTITY	Measurements, as of weight or distance
10	5	only two weeks	DATE	Absolute or relative dates or periods
11	6	Fralinger's	ORG	Companies, agencies, institutions, etc.
12	8	Wheatgrass	PERSON	People, including fictional
13	10	Tequila Picante Gourmet de Inclan	ORG	Companies, agencies, institutions, etc.
14	12	Felidae Platinum	ORG	Companies, agencies, institutions, etc.
15	12	more than two years	DATE	Absolute or relative dates or periods
16	13	Twizzlers	ORG	Companies, agencies, institutions, etc.

	Review_ID	Entity	Label	Description
17	14	The Strawberry Twizzlers	ORG	Companies, agencies, institutions, etc.
18	14	Six pounds	QUANTITY	Measurements, as of weight or distance
19	14	I.	ORG	Companies, agencies, institutions, etc.

```
label_counts = entities_df['Label'].value_counts().reindex(target_labels, fill_value=0)
display(label_counts.to_frame('Count'))

plt.figure(figsize=(9, 5))
ax = sns.barplot(x=label_counts.index, y=label_counts.values, hue=label_counts.index, palette='Set2', legend=False)
for container in ax.containers:
    ax.bar_label(container)
plt.title('Number of Entities per Label')
plt.xlabel('Entity Label')
plt.ylabel('Count')
plt.show()
```

	Count
Label	
PERSON	4431
ORG	6766
DATE	3107
MONEY	908
QUANTITY	680
PRODUCT	257




```
print("Top 10 most frequent entities for each label:")
top_entities = {}
for label in target_labels:
    top = entities_df[entities_df['Label'] == label]['Entity'].value_counts().head(10)
    top_entities[label] = [f"{e} ({c})" for e, c in top.items()] + [''] * (10 - len(top))
top_entities_df = pd.DataFrame(top_entities, index=range(1, 11))
display(top_entities_df)
```

Top 10 most frequent entities for each label:

	PERSON	ORG	DATE	MONEY	QUANTITY	PRODUCT
1	Keurig (214)	Amazon (1086)	years (107)	1 (26)	a ton (20)	Gatorade (18)
2	Switch (129)	Amazon.com (79)	Christmas (102)	10 (20)	34 grams (19)	Subscribe (15)
3	Bisquick (102)	Newman (70)	today (101)	2 (19)	1 cup (14)	Starbucks (8)
4	Newman (80)	Starbucks (58)	daily (84)	5 (19)	8.3 ounces (13)	CA (7)
5	Wolfgang Puck (61)	K-Cups (53)	the day (77)	20 (17)	12 ounce (11)	CO2 (5)
6	Bob (44)	BBQ (53)	a week (46)	3 (14)	16 ounces (10)	gooey (5)

	PERSON	ORG	DATE	MONEY	QUANTITY	PRODUCT
7	Pamela (40)	MSG (50)	a few days (33)	15 (11)	8.3 ounce (9)	Vanilla (5)
8	Chia (37)	Costco (43)	many years (27)	6 (11)	12 ounces (8)	Skittles (5)
9	Kettle (26)	Icicle (40)	the years (26)	every penny (10)	8 ounces (6)	Grove Square (4)
10	Wolfgang Puck's (25)	Wellness (37)	a month (25)	12 (9)	16-Ounce Boxes (6)	Butterfinger (4)

Information Extraction

Task 7: Function that returns tokens, lemmas, POS tags, named entities and entity labels for a review

```
def extract_information(review):
    doc = nlp(review)
    return {
        'Tokens': [token.text for token in doc],
        'Lemmas': [token.lemma_ for token in doc],
        'POS Tags': [token.pos_ for token in doc],
        'Named Entities': [ent.text for ent in doc.ents],
        'Entity Labels': [ent.label_ for ent in doc.ents],
    }
```

Task 8: Test the function with three new customer reviews

```
new_reviews = [
    "I ordered two pounds of Starbucks coffee from Amazon on January 5th and paid only $25. John from customer service was very
helpful!",
    "The Nutella jar arrived broken last Monday. I contacted Walmart support, but they refunded just 10 dollars for the 3 kg
pack.",
    "My daughter Emma loves these Kellogg's cereals. We buy 4 boxes every week from Target for about $18.",
]

extracted = [extract_information(r) for r in new_reviews]
for review, info in zip(new_reviews, extracted):
    print("Review:", review)
    for key, value in info.items():
        print(f"  {key}: {value}")
    print('-' * 110)
```

```
Review: I ordered two pounds of Starbucks coffee from Amazon on January 5th and paid only $25. John from customer service was very
helpful!
  Tokens: ['I', 'ordered', 'two', 'pounds', 'of', 'Starbucks', 'coffee', 'from', 'Amazon', 'on', 'January', '5th', 'and', 'paid',
'only', '$', '25', '.', 'John', 'from', 'customer', 'service', 'was', 'very', 'helpful', '!']
  Lemmas: ['I', 'order', 'two', 'pound', 'of', 'Starbucks', 'coffee', 'from', 'Amazon', 'on', 'January', '5th', 'and', 'pay',
'only', '$', '25', '.', 'John', 'from', 'customer', 'service', 'be', 'very', 'helpful', '!']
  POS Tags: ['PRON', 'VERB', 'NUM', 'NOUN', 'ADP', 'PROPN', 'NOUN', 'ADP', 'PROPN', 'ADP', 'PROPN', 'NOUN', 'CCONJ', 'VERB', 'ADV',
'SYM', 'NUM', 'PUNCT', 'PROPN', 'ADP', 'NOUN', 'NOUN', 'AUX', 'ADV', 'ADJ', 'PUNCT']
  Named Entities: ['two pounds', 'Starbucks', 'Amazon', 'January 5th', 'only $25', 'John']
  Entity Labels: ['QUANTITY', 'ORG', 'ORG', 'DATE', 'MONEY', 'PERSON']
--------------------------------------------------------------------------------------------------------------
Review: The Nutella jar arrived broken last Monday. I contacted Walmart support, but they refunded just 10 dollars for the 3 kg
pack.
  Tokens: ['The', 'Nutella', 'jar', 'arrived', 'broken', 'last', 'Monday', '.', 'I', 'contacted', 'Walmart', 'support', ',', 'but',
'they', 'refunded', 'just', '10', 'dollars', 'for', 'the', '3', 'kg', 'pack', '.']
  Lemmas: ['the', 'Nutella', 'jar', 'arrive', 'break', 'last', 'Monday', '.', 'I', 'contact', 'Walmart', 'support', ',', 'but',
'they', 'refund', 'just', '10', 'dollar', 'for', 'the', '3', 'kg', 'pack', '.']
  POS Tags: ['DET', 'PROPN', 'NOUN', 'VERB', 'VERB', 'ADJ', 'PROPN', 'PUNCT', 'PRON', 'VERB', 'PROPN', 'NOUN', 'PUNCT', 'CCONJ',
'PRON', 'VERB', 'ADV', 'NUM', 'NOUN', 'ADP', 'DET', 'NUM', 'X', 'NOUN', 'PUNCT']
  Named Entities: ['Nutella', 'last Monday', 'Walmart', 'just 10 dollars', '3 kg']
  Entity Labels: ['PERSON', 'DATE', 'ORG', 'MONEY', 'QUANTITY']
--------------------------------------------------------------------------------------------------------------
Review: My daughter Emma loves these Kellogg's cereals. We buy 4 boxes every week from Target for about $18.
  Tokens: ['My', 'daughter', 'Emma', 'loves', 'these', 'Kellogg', "'s", 'cereals', '.', 'We', 'buy', '4', 'boxes', 'every', 'week',
'from', 'Target', 'for', 'about', '$', '18', '.']
  Lemmas: ['my', 'daughter', 'Emma', 'love', 'these', 'Kellogg', "'s", 'cereal', '.', 'we', 'buy', '4', 'box', 'every', 'week',
'from', 'Target', 'for', 'about', '$', '18', '.']
  POS Tags: ['PRON', 'NOUN', 'PROPN', 'VERB', 'DET', 'PROPN', 'PART', 'NOUN', 'PUNCT', 'PRON', 'VERB', 'NUM', 'NOUN', 'DET', 'NOUN',
'ADP', 'PROPN', 'ADP', 'ADV', 'SYM', 'NUM', 'PUNCT']
  Named Entities: ['Emma', 'Kellogg', '4', 'every week', 'Target', 'about $18']
  Entity Labels: ['PERSON', 'ORG', 'CARDINAL', 'DATE', 'ORG', 'MONEY']
--------------------------------------------------------------------------------------------------------------
```

Task 9: Display the extracted information in a structured format

```
summary_df = pd.DataFrame({
    'Review': new_reviews,
    'Tokens': [', '.join(info['Tokens']) for info in extracted],
    'Lemmas': [', '.join(info['Lemmas']) for info in extracted],
    'POS Tags': [', '.join(info['POS Tags']) for info in extracted],
    'Named Entities': [', '.join(info['Named Entities']) for info in extracted],
    'Entity Labels': [', '.join(info['Entity Labels']) for info in extracted],
}, index=['Review 1', 'Review 2', 'Review 3'])
display(summary_df)
```

	Review	Tokens	Lemmas	POS Tags	Named Entities	Entity Labels
Review 1	I ordered two pounds of Starbucks coffee from Amazon on January 5th and paid only $25. John from customer service was very helpful!	I, ordered, two, pounds, of, Starbucks, coffee, from, Amazon, on, January, 5th, and, paid, only, $, 25, ., John, from, customer, service, was, ver...	I, order, two, pound, of, Starbucks, coffee, from, Amazon, on, January, 5th, and, pay, only, $, 25, ., John, from, customer, service, be, very, he...	PRON, VERB, NUM, NOUN, ADP, PROPN, NOUN, ADP, PROPN, ADP, PROPN, NOUN, CCONJ, VERB, ADV, SYM, NUM, PUNCT, PROPN, ADP, NOUN, NOUN, AUX, ADV, ADJ, P...	two pounds, Starbucks, Amazon, January 5th, only $25, John	QUANTITY, ORG, ORG, DATE, MONEY, PERSON
Review 2	The Nutella jar arrived broken last Monday. I contacted Walmart support, but they refunded just 10 dollars for the 3 kg pack.	The, Nutella, jar, arrived, broken, last, Monday, ., I, contacted, Walmart, support, ,, but, they, refunded, just, 10, dollars, for, the, 3, kg, p...	the, Nutella, jar, arrive, break, last, Monday, ., I, contact, Walmart, support, ,, but, they, refund, just, 10, dollar, for, the, 3, kg, pack, .	DET, PROPN, NOUN, VERB, VERB, ADJ, PROPN, PUNCT, PRON, VERB, PROPN, NOUN, PUNCT, CCONJ, PRON, VERB, ADV, NUM, NOUN, ADP, DET, NUM, X, NOUN, PUNCT	Nutella, last Monday, Walmart, just 10 dollars, 3 kg	PERSON, DATE, ORG, MONEY, QUANTITY
Review 3	My daughter Emma loves these Kellogg's cereals. We buy 4 boxes every week from Target for about $18.	My, daughter, Emma, loves, these, Kellogg, 's, cereals, ., We, buy, 4, boxes, every, week, from, Target, for, about, $, 18, .	my, daughter, Emma, love, these, Kellogg, 's, cereal, ., we, buy, 4, box, every, week, from, Target, for, about, $, 18, .	PRON, NOUN, PROPN, VERB, DET, PROPN, PART, NOUN, PUNCT, PRON, VERB, NUM, NOUN, DET, NOUN, ADP, PROPN, ADP, ADV, SYM, NUM, PUNCT	Emma, Kellogg, 4, every week, Target, about $18	PERSON, ORG, CARDINAL, DATE, ORG, MONEY

```
for i, info in enumerate(extracted, start=1):
    print(f"Review {i}: {new_reviews[i - 1]}")
    print("Token-level information:")
    display(pd.DataFrame({'Token': info['Tokens'], 'Lemma': info['Lemmas'], 'POS Tag': info['POS Tags']}))
    print("Named entities:")
    display(pd.DataFrame({'Entity': info['Named Entities'], 'Label': info['Entity Labels'],
                          'Description': [spacy.explain(l) for l in info['Entity Labels']]}))
    print('=' * 110)
```

```
Review 1: I ordered two pounds of Starbucks coffee from Amazon on January 5th and paid only $25. John from customer service was very helpful!
Token-level information:
```

	Token	Lemma	POS Tag
0	I	I	PRON
1	ordered	order	VERB
2	two	two	NUM
3	pounds	pound	NOUN
4	of	of	ADP
5	Starbucks	Starbucks	PROPN
6	coffee	coffee	NOUN
7	from	from	ADP
8	Amazon	Amazon	PROPN
9	on	on	ADP
10	January	January	PROPN
11	5th	5th	NOUN
12	and	and	CCONJ
13	paid	pay	VERB
14	only	only	ADV
15	$	$	SYM
16	25	25	NUM
17	.	.	PUNCT
18	John	John	PROPN
19	from	from	ADP
20	customer	customer	NOUN
21	service	service	NOUN
22	was	be	AUX
23	very	very	ADV
24	helpful	helpful	ADJ
25	!	!	PUNCT

```
Named entities:
```

	Entity	Label	Description
0	two pounds	QUANTITY	Measurements, as of weight or distance
1	Starbucks	ORG	Companies, agencies, institutions, etc.
2	Amazon	ORG	Companies, agencies, institutions, etc.
3	January 5th	DATE	Absolute or relative dates or periods

	Entity	Label	Description
4	only $25	MONEY	Monetary values, including unit
5	John	PERSON	People, including fictional

==========

```
Review 2: The Nutella jar arrived broken last Monday. I contacted Walmart support, but they refunded just 10 dollars for the 3 kg
pack.
Token-level information:
```

	Token	Lemma	POS Tag
0	The	the	DET
1	Nutella	Nutella	PROPN
2	jar	jar	NOUN
3	arrived	arrive	VERB
4	broken	break	VERB
5	last	last	ADJ
6	Monday	Monday	PROPN
7	.	.	PUNCT
8	I	I	PRON
9	contacted	contact	VERB
10	Walmart	Walmart	PROPN
11	support	support	NOUN
12	,	,	PUNCT
13	but	but	CCONJ
14	they	they	PRON
15	refunded	refund	VERB
16	just	just	ADV
17	10	10	NUM
18	dollars	dollar	NOUN
19	for	for	ADP
20	the	the	DET
21	3	3	NUM
22	kg	kg	X
23	pack	pack	NOUN
24	.	.	PUNCT

```
Named entities:
```

	Entity	Label	Description
0	Nutella	PERSON	People, including fictional
1	last Monday	DATE	Absolute or relative dates or periods
2	Walmart	ORG	Companies, agencies, institutions, etc.
3	just 10 dollars	MONEY	Monetary values, including unit
4	3 kg	QUANTITY	Measurements, as of weight or distance

==========

```
Review 3: My daughter Emma loves these Kellogg's cereals. We buy 4 boxes every week from Target for about $18.
Token-level information:
```

	Token	Lemma	POS Tag
0	My	my	PRON
1	daughter	daughter	NOUN
2	Emma	Emma	PROPN
3	loves	love	VERB
4	these	these	DET
5	Kellogg	Kellogg	PROPN
6	's	's	PART
7	cereals	cereal	NOUN
8	.	.	PUNCT
9	We	we	PRON
10	buy	buy	VERB
11	4	4	NUM
12	boxes	box	NOUN
13	every	every	DET
14	week	week	NOUN
15	from	from	ADP
16	Target	Target	PROPN
17	for	for	ADP

	Token	Lemma	POS Tag
18	about	about	ADV
19	$	$	SYM
20	18	18	NUM
21	.	.	PUNCT

```
Named entities:
```

	Entity	Label	Description
0	Emma	PERSON	People, including fictional
1	Kellogg	ORG	Companies, agencies, institutions, etc.
2	4	CARDINAL	Numerals that do not fall under another type
3	every week	DATE	Absolute or relative dates or periods
4	Target	ORG	Companies, agencies, institutions, etc.
5	about $18	MONEY	Monetary values, including unit

```
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```